

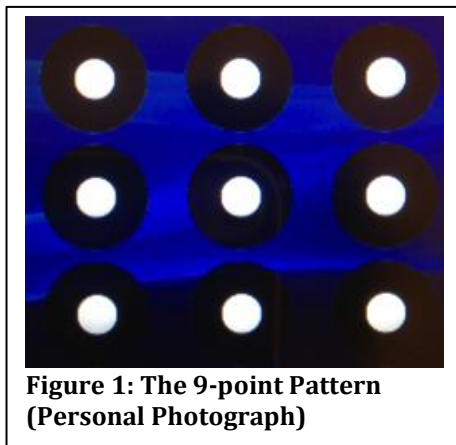
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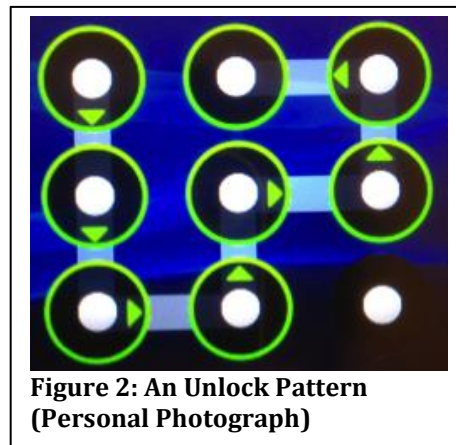
1. Introduction

The world is growing together through unimaginable amounts of information easily available on the Internet. At the same time, smartphones are increasingly becoming everyone's daily communications companions. Our smartphones manage our social contacts, e-mails, short messages, pictures and business communications. In times of smartphone theft, corporate espionage and cyber bullying, issues of data security become increasingly relevant. How can we ensure that our privacy and valuable information is secure on our smartphones? How can we ensure the safety of our device in the likely event of forgetting or losing it somewhere?

To address these issues, smartphone manufacturers have developed various lock screen systems aiming to safeguard access to the features and data on a smartphone. Before the full functionality of a phone woken from stand-by can be used, the user has to successfully enter the security code defined by the owner of the phone. The most popular variants of such systems are simple 4-digit numeric passcodes and the visual 9-point pattern unlock integrated into Google Android phone operating systems. The visual unlock system ensures the authenticity of the user by providing a set of 9 points arranged in square form on the touchscreen (as shown by Figure 1). The user is required to draw a pattern exactly matching the one pre-determined by the owner of the smartphone (an example is shown in Figure 2), before the smartphone unlocks. So how secure are such a visual lock screen codes really? Which of the two most common unlock systems, the 4-digit numeric code or the 9-point pattern unlock, provides greater security for concerned users?



**Figure 1: The 9-point Pattern
(Personal Photograph)**



**Figure 2: An Unlock Pattern
(Personal Photograph)**

As somebody who uses a smartphone on a daily basis, I encounter smartphone authentication systems often. Concerned about the security of my phone and the personal data on it, I wondered which of the system provides the highest number of unlock code possibilities, and thus, the highest level of security. Also, the common smudges on touchscreens originating from the oil on human fingers concerned me. I became curious, and wanted to understand the mathematical concepts that govern the number of possibilities of each unlock system.

1.1 Aims

This exploration aims to mathematically investigate the total number of pattern possibilities present within the visual 9-point unlock system, in order to compare its security to that of the standard 4-digit pin numbers otherwise used for authentication on smartphones. The investigation aims to achieve this with abstention from computerized brute-force methods, which systematically attempt all pattern possibilities by trial-and-error, thus investigating only the mathematical patterns and concepts that underlie 9-point and 4-digit unlock systems. In order to achieve this, the rules underlying the usability of 9-point and 4-digit unlock systems for the user will be analyzed. Introductions to permutations, sequences and series and graph theory will be

provided, and the concepts then applied to 9-point unlock patterns. Furthermore, the security risk posed by cases in which possible intruders obtain extra information from oily smudges or fingerprints on the touchscreen of a smartphone will be examined mathematically using a simplified consideration of the most common case. This is done, as the development of a generalized formula would strongly exceed the scope available for this investigation. A recommendation for smartphone users based on which system is more secure (in terms of the number of passcode/pattern possibilities) will be the final outcome of the investigation.

2. The Principles of 4-Digit Unlock Systems

Systems prompting the user to put in a 4-digit numerical passcode to unlock a smartphone are very common, and are built on the idea that the user has to know the exact code set by the owner of the phone in order to gain functional access. Any digit that lies on a phone number pad can be an *element*, or individual member, of the passcode *sequence*, the ordered arrangement of numbers required to unlock the phone (Ruohonen). As phone number pads only include 11 positive integers up to 9 and 0, all possible elements of the passcode must be an element of the set $X = \{a, a \in \mathbb{Z} \wedge 0 \leq a \leq 9\}$, meaning “the set of all a such that a is an element of the integers and larger than or equal to 0 and smaller than or equal to 9”. As any digit in the passcode can be used any number of times, each of the 4 different spots in the passcode can be filled with the number of elements in the set X , denoted by the *cardinality* of the set X :

$$|X| = 10$$

Therefore, for each of the ten possible digits in the first spot, 10 different digits can be chosen for the second spot, and for each of those ten, a further 10 can be chosen for the third spot, and so on. Therefore, the total number of distinct 4-digit unlock sequences that can be formed, b , is:

$$b = 10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = 10000$$

This number can later be used for comparison with the 9-point unlock pattern system.

2.1 An Analysis of Oily Smudges on 4-Digit Unlock Systems

As on any touchscreen, smudges and fingerprints can occur with 4-digit unlock systems, which may provide additional information about the passcode to any intruder. However, it must be noticed that fingerprints and smudges can be the result of any input to a smartphone’s touchscreen, and must not necessarily result from putting in the passcode. This random uncertainty underlies all smudges, even though some could possibly be eliminated due to their position on the screen not coinciding with any digits on the displayed number pad. Furthermore, as any digit can be used multiple times in the sequence, it is possible that one fingerprint represents multiple uses of a digit in the passcode.

As there are 10 different digits on the number pad, the number of prints on a screen that could theoretically originate from putting in the passcode, c must be an element of the set $Y = \{c, c \in \mathbb{Z} \wedge 0 \leq c \leq 10\}$. As the passcode is only four digits long, the total number of possible passcode sequences remaining with b fitting fingerprints on the screen, d , is:

$$d = b^4$$

However, this formula assumes *complete information*, or that no digit without a corresponding fingerprint can be part of the unlock sequence, which is not always the case and thus presenting another uncertainty underlying the use of smudges/fingerprints by intruders to crack passcodes. Without this simplification, the resulting uncertainty would increase the number of possibilities to the value stated in section 4, as all fingerprints could be unrelated to any passcode input.

3. The Principles of 9-Point Unlock Systems:

The Android 9-point unlock system is built on the idea that visual patterns are easier to remember for human beings than numeric codes, whilst still providing a great amount of unlock pattern possibilities to ensure security. The following principles govern the functionality of the software:

Firstly, the user is presented with the previously shown square of 3 by 3 points on the touchscreen. The individual points of the grid must serve as starting- or endpoints for the different line segments making up a visual unlock pattern. It is not possible for the user to input line segments starting or ending outside of the given grid. Furthermore, all line segments must be straight (grid criterion).

Secondly, any unlock pattern must be continuous, meaning that the line segments that make up a pattern can be drawn on the touchscreen without the fingertip having to leave the surface of the screen between line segments. Thus, all line segments of a pattern have to be consecutive to at least one other line segment part of the pattern (continuity criterion).

Also, line segments as part of the pattern can be drawn both between a point and those that are immediately adjacent to it (as shown in Figure 3), as well as points diagonally across from each other (shown by Figure 4), as long as no other point in the middle obstructs the direct path (adjacency criterion). However, once a point has already been part of a pattern, it can be “hopped across” (shown by Figure 5). This means that a line segment going over a point in between two points can now be drawn, if and only if the point in the middle is already part of the pattern (“hopper” criterion).

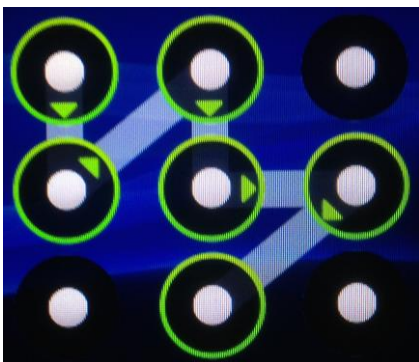


Figure 3: Lines Segments between directly adjacent Points (Personal Photograph)

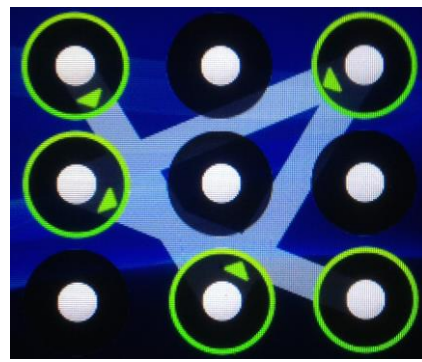


Figure 4: Pattern of long diagonal line segments (Personal Photograph)

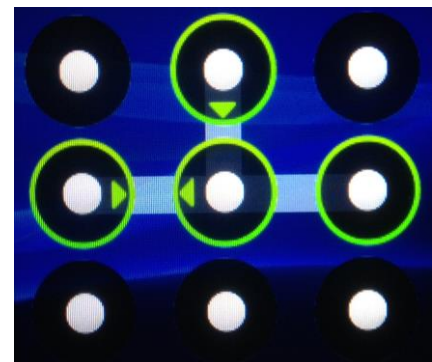


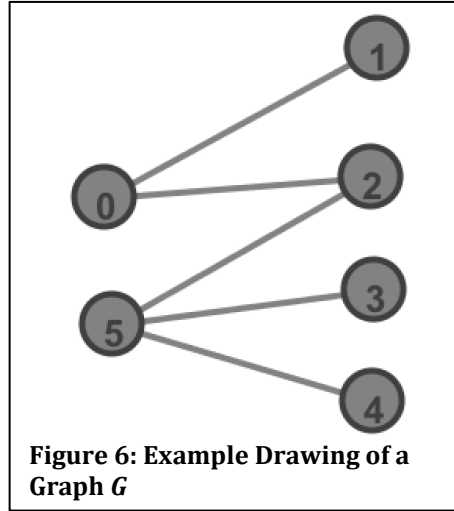
Figure 5: Pattern with a “hopper” segment over the center point (Personal Photograph)

Lastly, any unlock pattern must incorporate at least 4 distinctive points on the given grid joined by line segments. This is done to eliminate the possibility of insecure 2-point, single-line unlock patterns (length criterion). However, no point on the 3 by 3 point grid can be incorporated into an unlock pattern twice (singularity criterion). This limits the maximum number of points for pattern possibilities to a maximum of the 9 points.

4. An Introduction into Graph Theory:

Interestingly, when attempting to model visual unlock systems mathematically, the visual representations of 9-point unlock systems very much resemble the tools used in an area of mathematics known as graph theory. Graph theory is the mathematical study of relationships and networks, using visual tools known as *Graphs*, which represent mathematical sets of relationships between pre-determined variables or objects (Ruohonen). Figure 6 below shows an example

visual representation, also known as a *drawing*, of a graph G (Note that all graph drawings used in this investigation were created independently using the open-source software GraphTea).



This particular example visually shows a specific set of relationships between 6 different points, called *vertices*, each labeled as v_i , with i being an integer from 0 to 6 in this case. The lines that connect a pair of vertices, called *edges*, represent the relationships between them (Diestel). In the example graph G , vertex v_4 is connected to vertex v_5 by an edge. The relationships of graphs can also be expressed mathematically in the form of a *set of vertices* and a *set of edges*. Thus, graph G from above can be expressed as $G(V, E)$, where V is the set of vertices on the graph G and E its set of edges. Therefore for graph G :

$$V = \{ v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9 \}$$

$$E = \{ (v_0, v_1), (v_1, v_0), (v_0, v_2), (v_2, v_0), (v_2, v_5), (v_5, v_2), (v_3, v_5), (v_5, v_3), (v_4, v_5), (v_5, v_4) \}$$

Please note that every edge is denoted by the set of its vertex pair. Vertices that are joined by an edge are considered *adjacent vertices*. Edges that share exactly one common vertex are known as *adjacent edges*. Please note that the order within these vertex pairs does not matter for graph G , and each edge is thus denoted by two sets of vertices (Diestel). In this case, the graph characterized to be *undirected*, as the direction of a relationship between a pair of vertices does not matter (Diestel).

Graphs can also be *directed*, when the relationships between two vertices only exist in one direction, in which case the corresponding edge sets include the outgoing vertex, followed by the incident vertex. Edges on the drawing of a *digraph* are therefore associated with respective arrowheads at the incident vertex end of the edge (Ruohonen).

Vertices, can also be characterized by their *degree*, or the sum of edges that touch a vertex, denoted by $d(v)$ (Diestel). In example graph G , vertices v_1 , v_3 and v_4 are all of degree 1, vertices v_2 and v_0 are of degree 2 and vertex v_5 is of degree 3. Please note that every edge within a graph consists of a connection between two vertices and thus counts towards both vertices' degree once. Therefore, the total number of edges within a graph can be found as half of the sum of all vertex degrees. This concept can be expressed in sigma notation using the following formula, where n is the number of distinct vertices on the graph and m_{total} is its number of edges (Ruohonen):

$$m_{total} = \left(\frac{1}{2} \right) \cdot (d(v_1) + d(v_2) + d(v_3) + \dots + d(v_n)) = \frac{\sum_{l=1}^n d(v_l)}{2}$$

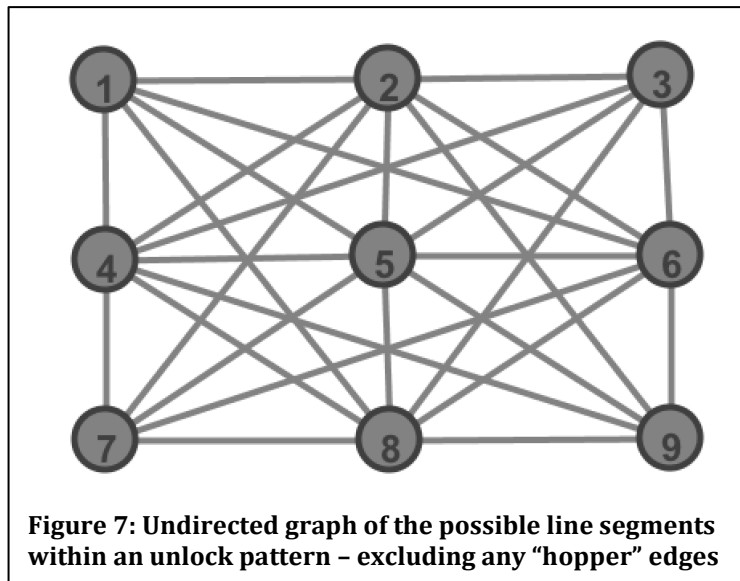
Note that sigma notation can be used to express the sum of a set of values by explicitly defining the set of values that is to be added using variables. In this case: “One half of the sum of all $d(v_l)$ from $l = 1$ up to $l = n$, where $l \in \mathbb{Z}$.”

Furthermore, every graph that is not a *null graph* (i.e. that does not consist of empty vertex and edge sets) or is not a graph containing only one vertex, can be considered consisting of several smaller disconnected components called *subgraphs* (Ruohonen). One type of subgraph, known as a *walk*, is defined as any route within a graph made up of consecutive adjacent edges, travelling over any edge or vertex any number of times. Another classification of subgraph, known as a *path*, is defined to be any walk that does not include any vertex more than once, with the exception that the starting and ending vertex may be the same. A path can thus also be either a *cycle*, when starting and ending vertex are the same, or a *trail*, when starting and ending vertices are different.

5. Analyzing 9-point Unlock Systems Mathematically:

Considering the principles and rules governing the 9-point unlock system, it can be said that any unlock pattern can be classified as a sequence of distinct adjacent edges (or vertices), in which each included edge is adjacent to no more than 2 other edges. Thus unlock patterns can mathematically be classified as trails. Furthermore, any unlock pattern must have a trail length of minimum 3 edges (minimum 4 vertices) and maximum 8 edges (all 9 vertices).

The set of possible edges that can be part of an unlock pattern can thus be represented in the form of a graph. Figure 7 shows all possible edges, excluding any “hopper” edges, as an undirected graph on the 3 by 3 point grid:



Note that figure 7 represents an undirected graph. Any edge on figure 7 represents two directed edges in opposite directions that could be part of an unlock pattern, thus the actual total number of possible edges is twice the number of edges on the graph shown.

Furthermore, looking at the degrees of the individual vertices on figure 7, a pattern can be observed. All vertices in the four corners of the grid $V_c = \{v_1, v_3, v_7, v_9\}$ are of degree 5. All vertices located in the middle of the outside borders of the grid $V_b = \{v_2, v_4, v_6, v_8\}$ are of degree 7. Only the vertex located in the middle of the grid v_5 is of degree 8. It can be noted that the differences in vertex degrees arise from the fact that some vertex pairs cannot be connected directly by a single, straight edge due to a blocking vertex in between. Table 1 lists the “blind spots”, or the vertices each vertex is not adjacent to, and classification of each vertex on the graph in figure 7:

Vertex	Vertex Type	Non-adjacent Vertices
v_1	Corner	v_3, v_7, v_9
v_2	Border	v_8
v_3	Corner	v_1, v_7, v_9
v_4	Border	v_6
v_5	Middle	-
v_6	Border	v_4
v_7	Corner	v_1, v_3, v_9
v_8	Border	v_2
v_9	Corner	v_1, v_3, v_7

Table 1: “Blind Spots” for all vertices

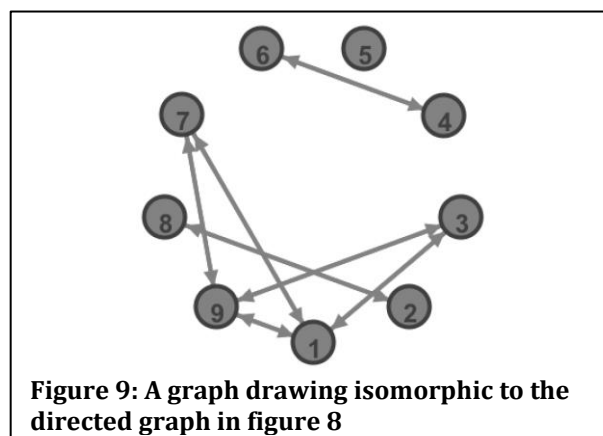
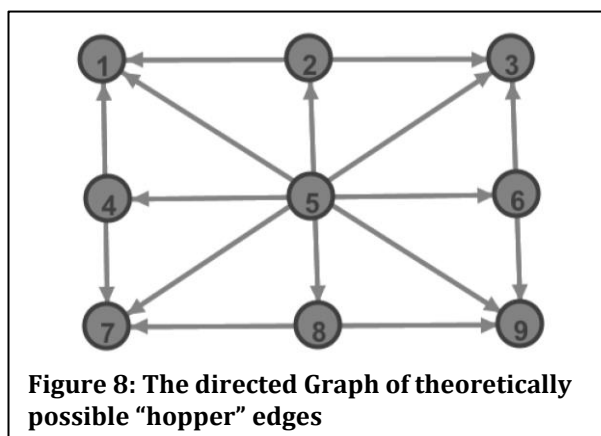
Upon examination of table 1, the following patterns can be recognized:

- Every corner vertex is only not adjacent to all other corner vertices, which are defined to be all odd-numbered vertices, excluding the considered corner vertex and the middle vertex v_5 .
- Any border vertex is not adjacent only to the border vertex directly opposite it on the grid, which is always defined as $v_{(10-i)}$ where i represents the considered border vertex
- The middle vertex v_5 is adjacent to all other vertices on the graph.

Considering the formula for the total number of edges explained in the previous section, the total number of edges that could be part of an unlock pattern (excluding “hopper” edges) can be calculated from figure 7. However, as every undirected edge in figure 7 represents two opposing directed edges, the result has to be multiplied by two:

$$m_{total} = \frac{\sum_{l=1}^n d(v_l)}{2} \cdot 2 = \sum_{l=1}^9 d(v_l) = 4 \cdot 5 + 4 \cdot 7 + 8 = 56$$

Theoretically, all the “hopper” edges can be included into the total number of possible individual edges forming an unlock pattern. However, it should be noted that these edges are only available for incorporation into an unlock pattern, if the point on the grid they jump across already is part of the pattern. In practice, this presents a considerable complication to calculating the total number of possible patterns. The directed graph shown in figure 8 can thus be drawn to display all theoretically possible “hopper” edges on the 3 by 3 point grid. Although the drawing of the graph in figure 9 may look as if all “hopper” edges are divided into two separate edges by the points they “jump”, this is not the case in practice. Vertex v_5 , for example, does not have any adjacent edges thus. For clarification, figure 9 shows an *isomorphic* graph to figure 8, or a different drawing of a graph representing the exact same set of vertices and edges (Weisstein).



Considering that each arrowhead represents the tip of one directed edge, it can be concluded that a total of 16 “hopper” edges could theoretically be part of an unlock pattern sequence by counting. Furthermore two cases of enabling “hopper” edges can be recognized. Looking at figure 8, it can be concluded that all members of the border set of vertices $V_b = \{v_2, v_4, v_6, v_8\}$ each enable 2 possible “hopper” edges across them if included. In contrast, the inclusion of the enabling vertex v_5 in a pattern sequence creates a total of 8 possible “hopper” edges available for selection later in the sequence.

Also, it can be noticed that the set of all possible “hopper” edges in figure 8 also represents the set of edges that is excluded from the graph in figure 7 due to non-adjacent vertices. Thus, if the two graphs from figures 7 and 8 were added, the resulting graph would be a *complete graph*, or a graph in which every theoretically possible vertex pair is connected by a unique edge (Ruohonen), on 9 vertices.

Thus the theoretical maximum number of possible edges available for selection, m_{max} , in a pattern sequence on the resulting complete graph is:

$$m_{max} = 56 + 16 = 72$$

Note that due to the continuity criterion, only edges adjacent to the previously selected edge are possible for selection. Therefore, the user has a choice of various possible edges or vertices for the consecutive step of the unlock pattern at each vertex. However the set of the edges remaining on the graph for possible inclusion in the pattern changes depending on the already selected edges due to the “hopper” and singularity criteria. Therefore the edge possibilities after each step in the pattern would have to be evaluated on a case-by-case basis. Trying to find the total number of distinct possible patterns using a flowchart will thus result in an irregular tree diagram. Furthermore, it was noticed that this tree diagram quickly explodes into thousands of branches within only 3 edges / 4 vertices. As the trail length of an unlock pattern can vary between 3 and 8 edges, 6 different tree diagrams would have to be constructed. This method of modeling would therefore represent a possible yet highly inefficient way of finding the total number of possible unlock patterns and was thus not pursued further.

However, as each unlock pattern can be expressed as an ordered sequence of vertices, the total number of pattern possibilities can also be found by finding the total number of vertex sequences that are possible with respect to the restrictions and principles of the unlock grid and software. Disregarding any restrictions to vertex sequences due to non-adjacent vertices, the total number of elements in the set of theoretically possible vertex sequences with a specific length r vertices, on the resultant complete graph without replacement (each vertex is selected only once per sequence), $|S_{tr}|$, can thus be found using *permutations*.

Permutations denote the total number of subsets with respect to order that can be found using r different elements of a set containing n total elements. Note that the total number of distinct vertices available for single inclusion in any sequence, $n = 9$. For $r = 9$, there are 9 vertices available to choose from to fill the first spot in the sequence, 8 for the second, and so on until only one is left. Thus the total number of arrangements for $n = 9$ and $r = 9$ is:

$$= {}_nP_r = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 9!$$

However, for $0 \leq r < 9$, the last $(n - r)$ spots of the sequence are not considered. Thus the number of arrangements that would otherwise originate from number of elements that can fill these spots has to be excluded by division.

$$= {}_nP_r = \frac{n!}{(n-r)!}$$

As $n = 9$ for all sequences of vertices, the total number of elements in the set of theoretically possible vertex sequences with length r , without replacement and without considering non-adjacent vertices can be described as:

$$|S_{tr}| = {}_nP_r = \frac{n!}{(n-r)!} = \frac{9!}{(9-r)!}$$

Considering that unlock patterns can have length $r = \{r \in \mathbb{Z} \wedge 4 \leq r \leq 9\}$, the total number of elements in the set of all possible vertex sequences, $|S_{t \text{ total}}|$, on the resultant complete graph is the sum of all possible $|S_{tr}|$:

$$|S_{t \text{ total}}| = \sum_{r=4}^9 \frac{9!}{(9-r)!} = 985824$$

Knowing $|S_{t \text{ total}}|$, the total number of elements in the set of patterns possible in practice, $|S_{p \text{ total}}|$, can be calculated by subtracting the total number of elements in the set of all practically impossible vertex sequences, $|S_{i \text{ total}}|$. As $\{S_{t \text{ total}}\}$ already complies with the singularity, continuity, length and grid criteria, only the vertex sequences not compliant with the adjacency and “hopper” criteria have to be considered for $\{S_{i \text{ total}}\}$. Consider the following process to express these two criteria mathematically to find $|S_{i \text{ total}}|$:

Firstly, the “hopper” and adjacency criteria can be expressed as a set of mathematical conditions in regards to the findings from the graph in figure 7 on page 6 as follows:

1. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_1, v_3\} \vee \{v_3, v_1\}$ and not containing the subset $\{v_2\}$ previously is $\in \{S_{i \text{ total}}\}$.
2. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_3, v_9\} \vee \{v_9, v_3\}$ and not containing the subset $\{v_6\}$ previously is $\in \{S_{i \text{ total}}\}$.
3. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_7, v_9\} \vee \{v_9, v_7\}$ and not containing the subset $\{v_8\}$ previously is $\in \{S_{i \text{ total}}\}$.
4. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_1, v_7\} \vee \{v_7, v_1\}$ and not containing the subset $\{v_4\}$ previously is $\in \{S_{i \text{ total}}\}$.
5. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_2, v_8\} \vee \{v_8, v_2\}$ and not containing the subset $\{v_5\}$ previously is $\in \{S_{i \text{ total}}\}$.
6. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_4, v_6\} \vee \{v_6, v_4\}$ and not containing the subset $\{v_5\}$ previously is $\in \{S_{i \text{ total}}\}$.
7. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_1, v_9\} \vee \{v_9, v_1\}$ and not containing the subset $\{v_5\}$ previously is $\in \{S_{i \text{ total}}\}$.
8. Any element of $\{S_{t \text{ total}}\}$ containing the ordered subsets $\{v_7, v_3\} \vee \{v_3, v_7\}$ and not containing the subset $\{v_5\}$ previously is $\in \{S_{i \text{ total}}\}$.

Studying these conditions, which exhaust all possible vertex sequence cases not possible in practice due to the “hopper” and adjacency criteria, a general formula for the number of distinct elements in the set, $|S_{i \text{ total}}|$, can be developed. Consider the following.

Within a sequence of length r containing the ordered subset such as $\{v_a, v_b\}$, the vertex pair of interest can take a number of distinct positions, q . Because the vertex pair has to stay in the specific consecutive order, it cannot be split up into two. Where u denotes the position of vertex within the sequence, with the first position being $u = 1$, the ordered pair of vertices must always

have positions (u) and $(u + 1)$. Considering that $(u + 1)$ can have a maximum value equal to r , the range of u within a vertex sequence of length r , is always equal to $(r - 1)$. As $u \in \mathbb{Z}^+$, the number of different u values possible for vertices in an ordered pair is equal to its range. As this number also represents the number of different positions the vertex pair can take within the sequence of length r , q , can be calculated as follows:

$$q = r - 1$$

For each possible value of u in a vertex sequence, a number of different vertex sequences can be formed. However, in order to eventually find $|S_{i \text{ total}}|$, only the total number of these possible vertex sequences in which the corresponding “hopper” edge-enabling vertex, v_e , occurs after the considered vertex pair should be calculated. As the vertices in the pair can take any u values within the possible range, the number of available spots in the sequence behind the vertex pair can range between 0 when $u = r - 1$ and $r - 2$ when $u = 1$. The exact number of spots available behind the vertex pair of each u value, w , can thus be calculated using the following formula:

$$w = r - u - 1$$

Thus the enabling vertex, v_e in this example, can take $(w + 1)$ different positions in order for the resulting sequence to be $\in \{S_{i \text{ total}}\}$ - it can take any spot behind (v_a, v_b) or not be included in the sequence at all (except for $r = n = 9$). For each distinct position of the enabling vertex behind the considered vertex pair, $r - 3$ spots in the sequence will remain available to be filled with the remaining $n - 3$ vertices. As $n = 9$ for all vertex sequences, the remaining number of available vertices to choose from will be 6. The number of different orders the remaining open spots can be filled with the available 6 vertices, o , can thus be calculated using the following formula. Please note that o therefore represents the number of possible vertex sequences of length r for one specific value of u and one specific position of v_e behind v_a, v_b .

$$o = {}_{(n-3)}P_{(r-3)} = {}_6P_{(r-3)} = \frac{6!}{(9-r)!}$$

As v_e can take w different positions behind (v_a, v_b) as well as not be included in the sequence at all, the number of elements of $\{S_{i \text{ total}}\}$ that can be formed using one specific vertex pair with corresponding enabling vertex and single specific values for u and $r \neq 9$ is:

$$o * (w + 1) = \left(\frac{6!}{(9-r)!} \right) \cdot (r - u)$$

However, for each vertex sequence of length r , u can take q different values ranging from 1 to $(r - 1)$. Thus the number of elements of $\{S_{i \text{ total}}\}$ that can be formed using one specific vertex pair with corresponding enabling vertex, and one specified value for r , where $r \neq 9$ is:

$$= \sum_{u=1}^{(r-1)} \left(\left(\frac{6!}{(9-r)!} \right) \cdot (r - u) \right)$$

As there are 16 sets of vertex pairs with a corresponding enabling vertex, and as the set of all possible values for r is defined as $\{4 \leq r \leq 9, r \in \mathbb{Z}\}$, the total number of elements of $\{S_{i \text{ total}}\}$, $|S_{i \text{ total}}|$ can be calculated. Note that $r = 9$ represents a special case where the number of spots v_e can take is equal to w , which is added to the number of possible vertex sequences of length $\{r \in \mathbb{Z} \wedge 4 \leq r \leq 9\}$. Therefore $|S_{i \text{ total}}|$ can be calculated using the following formula:

$$\begin{aligned}
|S_{i\ total}| &= \sum_{r=4}^8 \left(16 \cdot \sum_{u=1}^{(r-1)} \left(\left(\frac{6!}{(9-r)!} \right) \cdot (r-u) \right) \right) + 16 \cdot \sum_{u=1}^9 \left(\left(\frac{6!}{(9-9)!} \right) \cdot (9-u-1) \right) \\
&= \sum_{r=4}^8 \left(16 \cdot \sum_{u=1}^{(r-1)} \left(\left(\frac{6!}{(9-r)!} \right) \cdot (r-u) \right) \right) + 16 \cdot \sum_{u=1}^8 (6! \cdot (9-u-1))
\end{aligned}$$

Evaluating this formula using to find $|S_{i\ total}|$, the total number of impossible vertex sequences for 9-point unlock systems is:

$$|S_{i\ total}| = 788736$$

Now the total number of elements in the set of all possible vertex sequences in practice, $|S_{p\ total}|$ can be found by subtracting the number of impossible vertex sequences, $|S_{i\ total}|$, from the total number of theoretically possible vertex sequences, $|S_{t\ total}|$:

$$|S_{p\ total}| = |S_{t\ total}| - |S_{i\ total}| = 985824 - 788736 = 197088$$

Note that this result does not counts sequences that involve multiple illegal vertex combinations several times (once for each illegal vertex combination). The actual number of possible distinct pattern combinations is thus quite a bit higher – various different Internet sources suggest a value of 389112 obtained using brute-force methods (“Abstract Construct”), whose accuracy cannot be validated in this investigation, however.

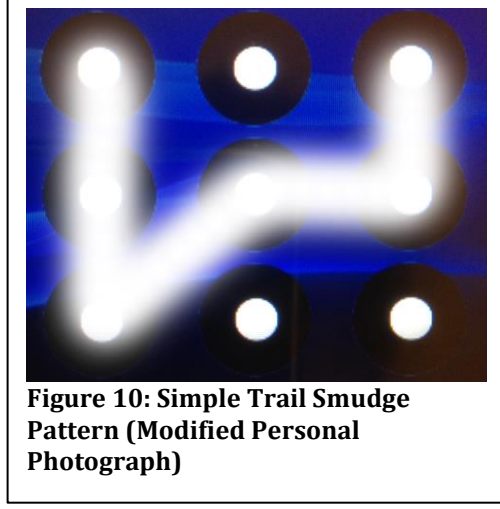
Several attempts were made to calculate the number of times that illegal vertex pairs were counted multiple times such that the result can be adjusted. However, no specific number could be calculated without individually considering each vertex sequence contained within $\{S_{t\ total}\}$, a process that would exceed the resources and boundaries of this investigation. This phenomenon occurs due to exploding amount of factors that have to be considered with multiple illegal vertex pairs. A vertex within one illegal pair can be the enabling vertex that makes a subsequent pair legal. Two different sequential vertex pairs can contain a common vertex contained in both, and various different spots for their distinct enabling vertices may be possible – whose arrangement may create another illegal vertex pair.

Thus, the conclusion of this section lies in knowing that at least 197088 different legal pattern possibilities exist for 9-point visual unlock systems, and that no formulaic way of calculating the total number of possibilities could be found in this case, without resorting to a brute-force-like trial-and-error examination of all theoretically possible vertex sequences.

5.1 Examining the Case of Oily Smudges on 9-Point Pattern Unlock:

Just as considered in section 4 for 4-digit passcodes, oily smudges may limit the number of different visual pattern possibilities and thus make it easier for possible intruders to “guess” the user’s pattern. However, a variety of different kinds of visual unlock pattern smudges (circuit smudges, smudges from patterns with overlaps, simple trail smudges) can withhold a variety of unlock patterns.

As I have informally found simple trail smudges including between 5 and 7 vertices to be the most common type of unlock pattern used among my friends, such a case will be analyzed. Note that the findings of this survey are in no way practically generalizable, as survey was conducted informally for the sole purpose of deciding on one common smudge pattern for analysis. The 6-vertex simple trail smudge case that will be considered is shown in figure 10 below.



Furthermore, as was considered for fingerprints on 4-digit passcodes, complete information will be assumed in this case, meaning that no actual edge of the unlock pattern can exist without a smudge relating to it, but that a smudged edge can exist on the screen without relating to an edge that is actually part of the pattern. This simplification is assumed as otherwise, all smudge edges could be completely unrelated to the actual unlock sequence – a case that is always a possibility in practice, where the information provided by the smudges is irrelevant and the number of possible pattern possibilities would be equal to that total number of possible patterns considered in section 5.

Considering these restrictions and the case displayed in figure 10, the number of different unlock patterns that could be inscribed within the 6-vertex trail smudge, g , can be calculated using the following pieces of information obtained from the smudge. Firstly, the intruder can exclude any possible vertex sequences of lengths less than 4 vertices, as they do not meet the length criterion preset by the software. Secondly, any intruder can be certain that any of the 6 vertices included by the smudge could also be part of the pattern. Thirdly, due to the simplification assumed above, any of these 6 vertices could only be directly adjacent to the vertices it is directly linked to by the smudge. Lastly, the smudges can be seen as the edges of an undirected graph, as the smudges do not provide information on which direction the edges of a pattern were entered. Therefore, considering the vertex naming conventions used in figure 7, the intruder knows that the actual unlock pattern must be inscribed in one of the two following vertex sequences S_1 and S_2 :

$$S_1 = v_1, v_4, v_7, v_5, v_6, v_3$$

$$S_2 = v_3, v_6, v_5, v_7, v_4, v_1$$

As $4 \leq r \leq 6 \wedge r \in \mathbb{Z}$, where r represents the length of the actual pattern within the smudge pattern, the number of patterns that can be inscribed in the above sequences will be the sum of all inscribable vertex sequences of lengths 4, 5 and 6 vertices. Considering h to be the number of the position of the first vertex of a sequence of length r inscribed within S_1 or S_2 , the minimum and maximum values for h can be found as:

$$h_{min} = 1$$

$$h_{max} = 6 + h_{min} - r = 7 - r$$

This is due to the fact that only one sequence of length 6 can be inscribed into another sequence of length 6, whereas two different sequences of length 5 can be inscribed into the same original sequence of length 6. Thus the number of different inscribable sequences, j , of a defined length r ,

can be found by calculating the range of all possible values for h and adding one (for the inscribed sequence obtained when $(h = 1)$).

$$j = h_{max} - h_{min} + 1$$

Therefore, the total number of sequences that can be inscribed into another sequence of length 6 can thus be described as the sum of all ranges of h . As there two different vertex sequences into which other sequences can be inscribed, the total number of patterns that is inscribed within the smudge in figure 10, g , assuming the stated simplifications is:

$$g = 2 \cdot \sum_{r=4}^6 (7 - r - 1 + 1) = 2 \cdot \sum_{r=4}^6 (7 - r) = 12$$

6. Comparison of Smartphone Unlock System Security

Using the findings from sections 2, 2.1, 5 and 5.1, a numeric comparison between the two unlock systems can be done. Firstly, it can be said that the number of distinct unlock sequences that could theoretically be set by the user of a smartphone is much higher for the 9-point visual unlock systems than for 4-digit numeric passcodes, even if the exact number of distinct pattern possibilities available for 9-point unlock could not be found exactly, as:

$$197088 > 10000$$

is a true statement. Therefore, assuming an intruder has no other information available, 9-point unlock systems can be considered more secure. It will take an intruder much longer to try out all possible sequence possibilities (and thus definitely discover the pattern set by the smartphone owner) with 9-point unlock than with a 4-digit passcode. However, the exact probability of an intruder being able to “crack” the owner’s code is dependent on a phone’s operating system version, as different operating systems allow various different number of failed unlock sequence attempts before the smartphone goes into “lockdown” mode. Then, it either has to be connected to the computer of the owner, or no unlock attempts can be made within a certain amount of time, which increases between every failed unlock attempt.

Secondly, considering cases of smudges or fingerprints on touchscreens that provide extra information (assuming a complete information simplification), a common smudge on 9-point unlock provides surprisingly few possibilities. Equating a the expression found in section 2.1 with the number of sequence possibilities found in section 5.1, and solving it for b can be done to find the maximum number of distinct fingerprints required to make the remaining passcode possibilities on 4-digit unlock smaller than the number of possible unlock sequences within the pattern unlock smudge considered in section 5.1.

$$b^4 = 12$$

$$b = \sqrt[4]{12} = 1.86$$

However, as $b \in \mathbb{Z}^+$, the maximum number of fingerprints for 4-digit unlock to provide less pattern possibilities than inscribed within the smudge from section 5.1 is the next lower whole integer, $b = 1$. Depending on how the user stores his smartphone throughout the day, the occurrence of smudges or fingerprints may be more or less frequent. Users keeping it in a pocket of their pants will encounter smudges/prints less often, because the cloth of the pants repeatedly wipes away the smudges/prints on the touchscreen. Users keeping their smartphone in a handbag or backpack may thus have smudges on their touchscreens more often, making them more vulnerable to giving away information to possible intruders.

7. Conclusions:

Considering the results of this investigation, several conclusions about the security of 9-point unlock systems in comparison to 4-digit unlock systems can be made. Firstly, it can be said that 9-point unlock can be modeled using graph theory. However, the deductive approach to calculating the total number of visual unlock pattern possibilities that originated from this model was not successful as vertex sequences including several illegal vertex pairs were subtracted twice or more from the total number of vertex sequences without restrictions. Any attempts at adjusting the final result for this case resulted in using either a brute-force like examination of all vertex sequences without restrictions or having to create an irregular tree diagram – both presenting methods that would exceed the resources and boundaries of this investigation. Thus also the value various Internet sources suggest using brute-force methods cannot be validated.

Secondly, it can be said that while 4-digit unlock has 10000 different passcode possibilities, 9-point visual systems have at least 197088 different possible vertex sequences. 9-point visual unlock systems are therefore more secure, assuming an intruder has no other information about a user's unlock code.

Thirdly, considering the two different smudge/fingerprint cases examined in this investigation, it can be said that if the user of a smartphone frequently encounters smudges on his touchscreen, 4-digit unlock systems are likely to provide more security for users. Thus this unlock system is recommendable in practice for these smartphone users. For users who rarely encounter smudges or fingerprints on their smartphone screen, the use of 9-point pattern unlock is recommended as the number of patterns any intruder would have to try out, assuming no other information, is much higher. Knowing this, it is possible for various smartphone users to adapt the unlock system they use to the kind of smartphone user they are, in order to ensure maximum security for personal data, contacts, or corporate information.

Note that several limitations underlie the application of the findings of this investigation, as several assumptions were made in the process; all of which should be considered in association with the findings and their implications. See section 8 for details.

Overall, this was a very interesting yet often challenging investigation. Right from the beginning, my enthusiasm for puzzle solving sparked. I soon found the process of investigating and mathematically modeling visual unlock patterns quite a bit more substantial than originally expected. Furthermore, it was very interesting to explore the basics of Graph Theory in the process: an area of mathematics of which I had not been aware previously.

8. Limitations & Outlook:

However, several limitations underlie this investigation's findings. Considering that this investigation was not able to calculate a concrete value for the total number of pattern possibilities for 9-point unlock systems, it can only be said that the number of pattern possibilities is definitely more than the calculated value. Thus an exact numerical comparison quantifying the differences in security between 4-digit unlock and 9-point unlock is not possible.

It remains open whether 9-point visual unlock systems can be completely and accurately modeled using graph and set theory, or whether brute force methods represent the only methodology to find the total number of pattern possibilities. Further investigations into the mathematics behind vertex sequences including several illegal vertex pairs, and into the mathematics behind dynamic systems such as 9-point pattern unlock in the future can help further understand this limitation. Lastly, as digital technology, high-power computing and brute-force "cracking" of codes become

more common on a global scale, it remains uncertain to what extent smartphone unlock systems are actually secure (in general – from an absolute standpoint), and to what extent thus the difference in total pattern/passcode possibilities is practically relevant in regards to the use of such fast and aggressive hacking methods.

Also the simplification assuming complete information when considering cases of smudges or fingerprints on touchscreens is not always the case in reality. Often, a large uncertainty regarding the relevance of smudges or fingerprints in regards to the input of unlock codes remains. However, this is also dependent on the amount of information an intruder has available. An intruder may know the owner of a smartphone personally, and may have observed him or her entering the passcode – severely decreasing the number of patterns the intruder has to try before finding the correct one. These additional factors of uncertainty should be taken into account to evaluate and compare the security of 4-digit and 9-point unlock systems more precisely.

Only one case of smudges was analyzed for 9-point unlock systems. This further restricts the reliability of the results, as other types of smudges are likely in practice. This issue presents the opportunity of a large-scale study focusing on the types of unlock patterns frequently chosen, and the frequency of different kinds of smudges that result, to accurately evaluate the security and relevance of smudges for 9-point unlock patterns. The results of a small-scale investigation found on the Internet have also shown that it is likely that most people do not use 9-point unlock patterns to their full complexity, choosing simple and short vertex sequences. Furthermore, it is suggested that certain vertices may be more popular as starting or ending vertices of unlock patterns – circumstances that should be considered as they may limit the security of 9-point unlock in practice (Uellenbeck et. al.).

Lastly, several different other types of unlock systems not considered in this investigation are currently becoming more widespread, such as biometric fingerprint-based systems or facial recognition systems. These systems may be more secure than 4-digit or 9-point unlock systems, and should be considered when making recommendations for the most secure screen lock methodology.

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